

COMP20008: Elements of Data Processing, A2: Project 6

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1. Executive Summary

This study seeks to elucidate the complex elements that determine the popularity of movies and television shows, with a strong focus on what combinations yield good IMDb and TMDB scores. For both categorical and numerical features, our data preprocessing includes the imputation of missing values using K-Nearest Neighbors (KNN) and text preprocessing techniques like tokenization and lemmatization. Label encoding also translated categorical data into numerical form for analysis.

Two different modelling methodologies have been used in this paper. To begin, supervised regression models were built to predict IMDb and TMDB scores, but they had mediocre predictive performance with low R-squared values. Second, unsupervised PCA models were used to identify the underlying patterns in the data, which revealed that the first few main components explained a considerable portion of the variance.

The PCA analysis demonstrates the significance of runtime, IMDb votes, TMDB popularity, and age certification in shaping the variance for the entirety of the dataset. While the regression models provide information about predicted accuracy, the PCA models provide information about feature relevance and the inherent structure of the data.

2. Research Question

To analyse the factors influencing movie/show popularity based on IMDb score and TMDB score.

3. Dataset Explorations

Exploring credits.csv:

The dataset "credits.csv" is an important component of the research that aims to analyse the elements influencing the popularity of movies or series based on IMDb and TMDB data. This data set has cast and crew information for each of the show and movie titles. Each row in this data represents a person and has the columns person_id, name, character (if applicable), and role (e.g., ACTOR, DIRECTOR).

The dataset will be used in this study to explore the impact of key persons, mainly actors and directors, on the popularity of a film or television show. It enables the investigation of whether the presence of prominent performers such as Robert De Niro and Jodie Foster, or esteemed directors such as Martin Scorsese, has a substantial influence on a production's reception on platforms such as IMDb and TMDB.

Furthermore, this dataset allows for the evaluation of characteristics such as an actor's number of roles played, the frequency of collaborations between actors and directors, and whether uncredited roles have any influence on a production's popularity.

Exploring titles.csv:

The file "titles.csv" is the most important piece of data in this entire report, as it has all the elements impacting the popularity of movies and series based on IMDb and TMDB data. This dataset provides an in-depth look at various titles, including movies and TV shows, with attributes such as title, type (MOVIE/SHOW), description, release year, age certification, runtime, genres, production countries, seasons (for shows), IMDb ID, IMDb score, IMDb votes, TMDB popularity, and TMDB score.

"titles.csv" allows us to expand on the aspects of every movie. For starters, it makes it easier to assess the impact of numerous variables on a title's popularity, such as genre, runtime, release year, and age certification.

Furthermore, the dataset enables us to compare aspects of movies and shows to determine whether there are distinct reasons driving the popularity of these two types of material.

4. Data Pre-processing

In the zip file submitted, our preprocessing code is in the preprocessing.py file. Kindly refer to the README file for detailed instructions on how to run the code.

The four data preprocessing techniques that we decided to use in order to work with titles.csv and credits.csv data are as follows:

- 1) Data Imputation
- 2) Text Preprocessing
- 3) Label Encoding
- 4) Discretization

In this section, we expand further on the functions written in the preprocessing.py file.

1. Data Imputation:

In the function impute_age, missing values in the 'age_certification' column are imputed using the K-Nearest Neighbors (KNN) imputation method. The missing values in this categorical column are replaced with values based on their similarity to other samples using KNN.

In the functions impute_votes_popularity and impute_scores, missing values in the columns 'imdb_votes', 'tmdb_popularity', 'imdb_score', and 'tmdb_score' are imputed using KNN imputation. These numerical values are critical in sections 7 & 8

2. Text Preprocessing (Data Manipulation):

We performed text preprocessing on textual attributes such as 'title' and 'description'. It converts text to lowercase, removes punctuation, tokenizes text, removes stopwords, and lemmatizes words. We basically create a bag of words representation through the use of tokenization.

This transformation converts text into numerical features, making it suitable for machine learning algorithms.

3. Label Encoding:

In the encode_age_cert function, the 'age_certification' column, which contains categorical labels, is encoded into numerical values. This encoding allows categorical data to be used in numerical-based algorithms and analyses. Age certifications are mapped to numerical values representing different levels of maturity ratings.

4. Discretization:

The discretized_release function takes the “release_year” column as an input and sorts them into three categories: “OLD” (-2010), “MEDIUM” (2011-2019) and “NEW” (2020-), according to a titles release year. The years were chosen such that each category had roughly the same number of titles in each for easy analysis.

5. Pre-processing Analysis

To obtain a grasp of the elements impacting movie popularity, particularly in terms of IMDb and TMDb scores, our research team conducted a meticulous pre-processing analysis. This project aimed to explore the complex relationships between many elements and the IMDb votes received by movies. We created three distinct figures to shed light on this, each of which provides unique insights into the dynamics of movie popularity.

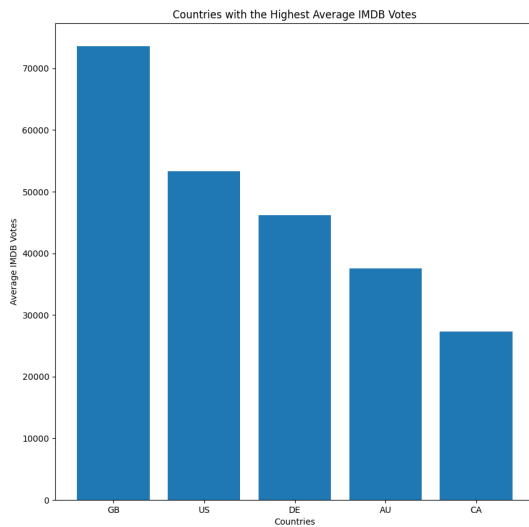


Figure 1 - Bar graph of countries with the highest average IMDb votes

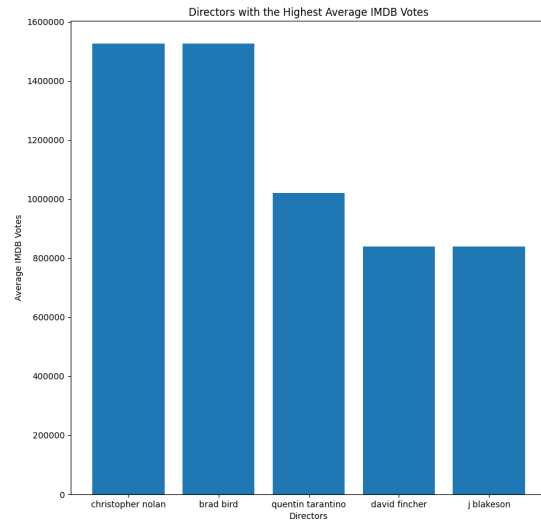


Figure 2 - Bar graph of directors with the highest average IMDb votes

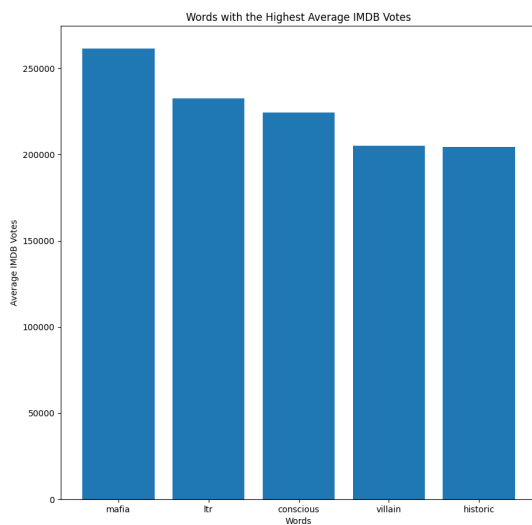


Figure 3 - Bar graph of words in descriptions with the highest average IMDb votes

Figure 1: Impact of Country Groupings on IMDb Votes

Our investigation began with the production of Figure 1, in which we attempted to analyse the impact of various parameters on IMDb votes. Notably, Figure 1 indicated a strong trend: movies from certain groups of countries received more average IMDb votes than movies from individual countries. This tells us that country clusters for movies are a very real factor. Since certain country regions have similar cultures, movies from nearby regions are both applicable in context and language, especially in these times where subbing and dubbing of movies and shows is commonplace. Following that, we updated our approach to show each country individually in the graph, even if they were part of a group.

Figure 2: Director Influence on Movie Popularity

Figure 2 investigated the influence of directors on movie popularity and discovered an interesting phenomenon. The notion of “one hit wonders”. Some directors were in the top popularity results because of a single successful movie. This gave us misrepresented data in our director influence analysis. To improve the accuracy of our analysis, we decided to include only filmmakers who had worked on two or more films, resulting in a more representative graph. Additionally, we expanded our investigation to include all directors in the movie dataset. We can see clearly that there is a sudden drop off in the average number of votes received by directors in figure 2. This could be due to national or personal popularity, global recognition, marketing schemes etc.

Figure 3: Description-Driven Popularity Insights

In Figure 3, we turned our attention to the textual characteristics of movie descriptions and their significance in predicting popularity. We found something very interesting regarding the word threshold for descriptions and titles. Setting a low criterion, or none at all, resulted in the inclusion of words that may have featured only once in the description of highly rated movies, with negligible influence on film popularity. As a result, we chose a threshold of 20, limiting our study to words that appeared 20 or more times in movie descriptions. If this is the case, we believe it is a popular word to use for describing movies, and hence gives a better representation of the highest average imdb votes for description words.

6. Dataset Limitations -> Pre-processing

1. Age Certification Imputation: The imputation of missing values in the 'age_certification' column was based on genre and description information. However, due to runtime considerations and the idea that the title may not have a major impact on age certification, the decision was made to omit the title from determining age certification imputation. Additionally, in the preprocessing stage of “description”, non-useful strings were created, creating noise for analysis.
2. Runtime Computation: Age certification based on runtime is computationally expensive, resulting in large delays (35 minutes) even when running on a 8GB optimised Macbook system. This limits the amount of information that can be extracted in short amounts of time, and we might have to work with a subset of the data to predict the entire data if there was a wall time limit.
3. Title Influence: Initially, a 'titles_array' comprising unique words from titles was included, but its impact to age certification prediction was not the next to say the least. As a result, it was omitted from the analysis, raising questions regarding the extent to which movie titles influence age certification. For example, the R rating of "Taxi Driver" does not indicate age classification. Since the words “taxi” and “driver” does not indicate that it is a crime film and is hence rated R.
4. Number of nearest neighbours for KNNImputer was chosen arbitrarily, and may not be chosen optimally in relation to the dataset size. In future, the elbow method or similar algorithm could be utilised in order to find a more mathematically optimal ‘n’ for imputation.
5. Misclassification of Documentaries: Documentaries are categorised as "SHOWS" in the dataset; however, they do not usually follow the same trend as shows. Documentaries are often single-season films with extended runtimes, and may follow trends closer to movies. Further analysis is required to address this.

7. Supervised Model

Introduction:

The goal of this supervised regression model is to look into the elements that influence the popularity of movies and TV shows based on their IMDb and TMDB scores. To forecast these scores, the model considers factors such as runtime, age certification, IMDb votes, and TMDB popularity. We can acquire insights into what drives the popularity of movies and shows by analysing the link between these features and the scores.

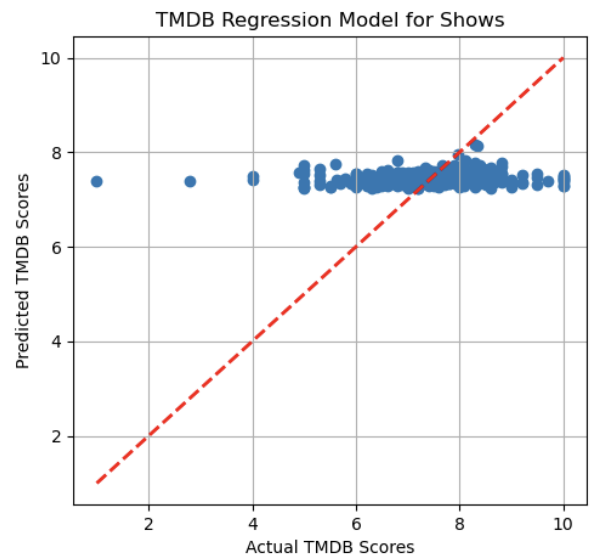
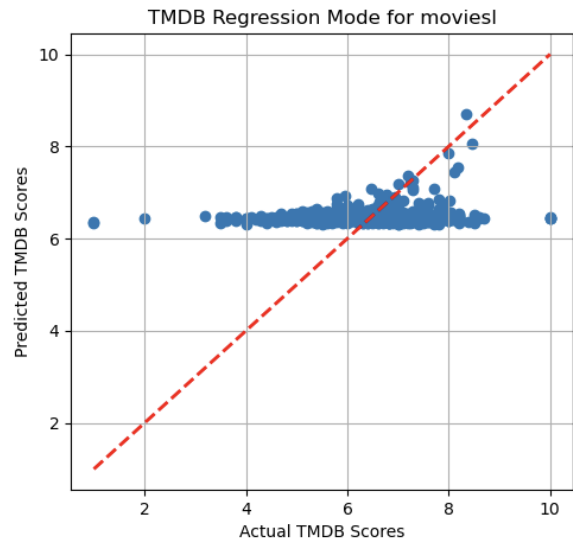
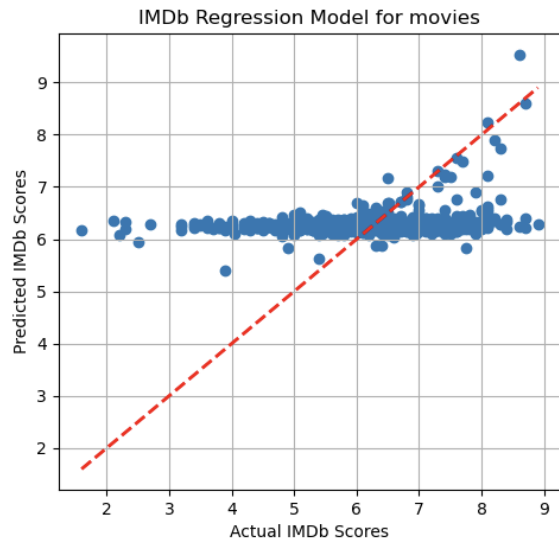
Core Components of the Model:

- **Data Loading and Splitting:** The model loads data from “movies.csv” and “shows.csv”, identifies relevant features and target variables for IMDb and TMDB scores, then divides the data into training and testing sets.
- **Standardisation:** It uses StandardScaler to achieve consistent scaling for both the IMDb and TMDB models.
- **Linear Regression Models:** To establish the correlations between features and scores, linear regression models are built for both IMDb and TMDB scores.
- **Model Evaluation:** For both IMDb and TMDB models, the model analyses performance using mean squared error (MSE) and R-squared (R2), providing insights into predicted accuracy.

OUTPUT

-----supervised movies model output----- IMDB Model Metrics for Movies: Mean Squared Error (MSE): 1.23 R-squared (R2): 0.05 TMDB Model Metrics for Movies: Mean Squared Error (MSE): 1.04 R-squared (R2): 0.03	-----supervised shows model output----- IMDB Model Metrics for Shows: Mean Squared Error (MSE): 0.90 R-squared (R2): 0.09 TMDB Model Metrics for Shows: Mean Squared Error (MSE): 1.07 R-squared (R2): 0.02
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Visualisations:



Findings and Analysis:

The model provides crucial insights about the popularity of movies and television shows. It implies that the number of seasons has a considerable impact on show scores, implying that highly rated series tend to have more seasons. Furthermore, IMDb votes appear as an important component for both movies and TV shows, indicating that well-received content gains traction and popularity. However, predicting scores is difficult for the model because they are extremely subjective and not always identifiable from numerical data. Overall, the findings emphasise the complex character of popularity and the importance of audience involvement and consistent quality in the entertainment sector.

The output metrics of the model provide a quantitative assessment of its predicted performance. Both the IMDb and TMDB models for movies have weak predictive accuracy, with low R-squared values. This implies that the selected features, such as runtime, age certification, and IMDb votes, explain just a small percentage of the variation in movie popularity. Similarly, the models for TV series demonstrate modest predictive power, with slightly higher R-squared values for IMDb scores. These findings support the notion that the factors determining movie and television show popularity are complex and extend beyond the attributes included in this model.

Limitations:

- The model is based on a limited collection of features, eliminating potentially relevant criteria such as cast, genre, and release date.
- Data Quality: The model's performance is determined by the dataset's quality and representativeness. Biases or missing data could have an impact on the outcomes.
- Overfitting: If the dataset is small or there is multicollinearity across characteristics, linear regression may fail to generalize well.

Possible Future Improvements:

Since the present models have low predictive ability (low R-squared values), future upgrades to this model could include advanced regression approaches (non-linear) or incorporating more relevant features based on findings. Testing the model on other datasets that are larger or have higher data integrity in order to better train the model.

8. Unsupervised Model

Introduction:

The goal of this unsupervised PCA (Principal Component Analysis) model is to look into the elements that influence the popularity of movies and TV shows based on IMDb and TMDB scores. This analysis provides insights into how factors like runtime, IMDb votes, TMDB popularity, and age certification contribute to the total popularity of movies and series by lowering the dimensionality of the dataset and visualising feature loadings in principal components. The scree plot and cumulative explained variance plot show the importance of each principal component in explaining the variation in the data, which aids in feature selection and recognising the most relevant aspects.

Core Components of the Model:

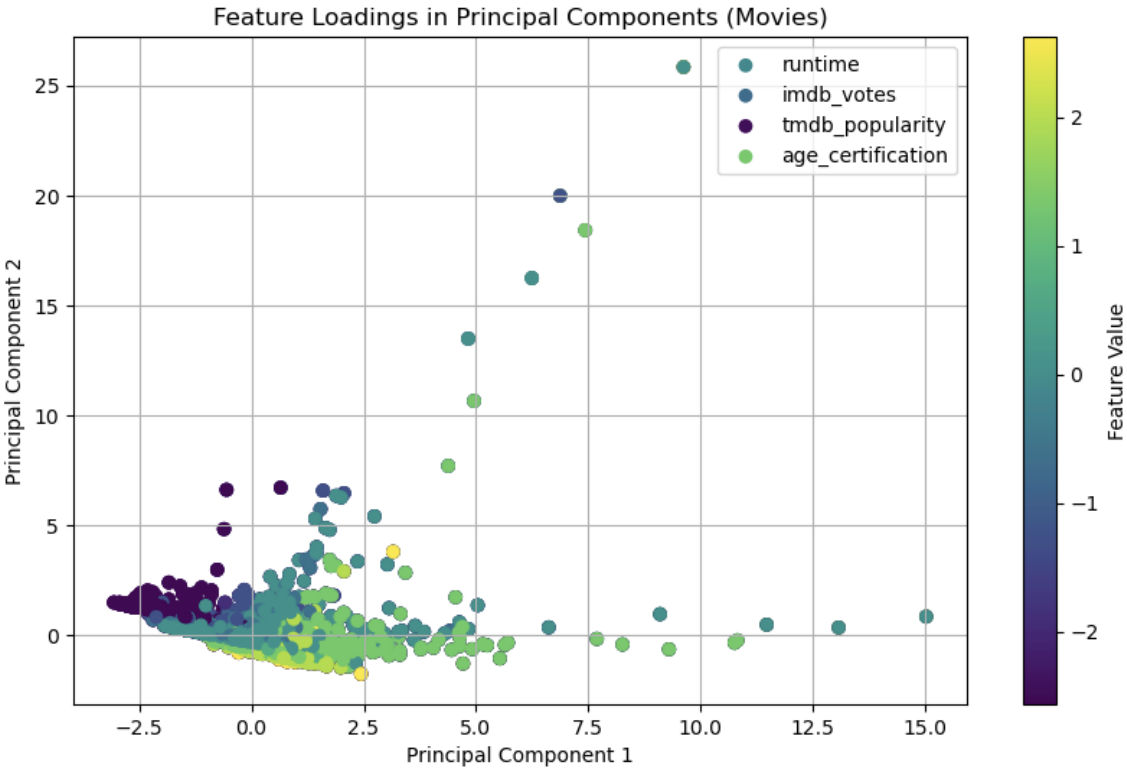
- Standardisation: To ensure uniform scaling and to prevent any single feature from dominating the study, standardise the selected features.
- Principal Component Analysis Dimensionality Reduction: Use Principal Component Analysis to reduce data dimensionality while keeping key variance.
- Explained Variance Analysis: Determine the number of primary components required to capture a significant amount of the data's variability using the explained variance ratio and cumulative explained variance.

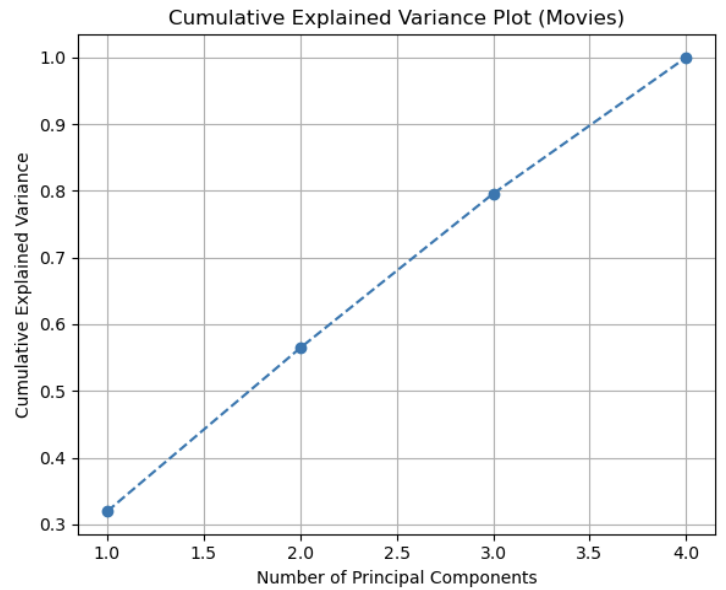
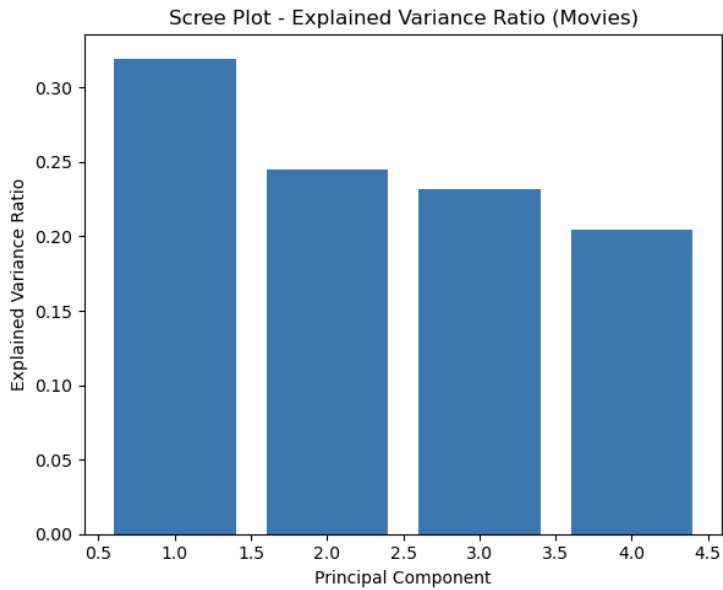
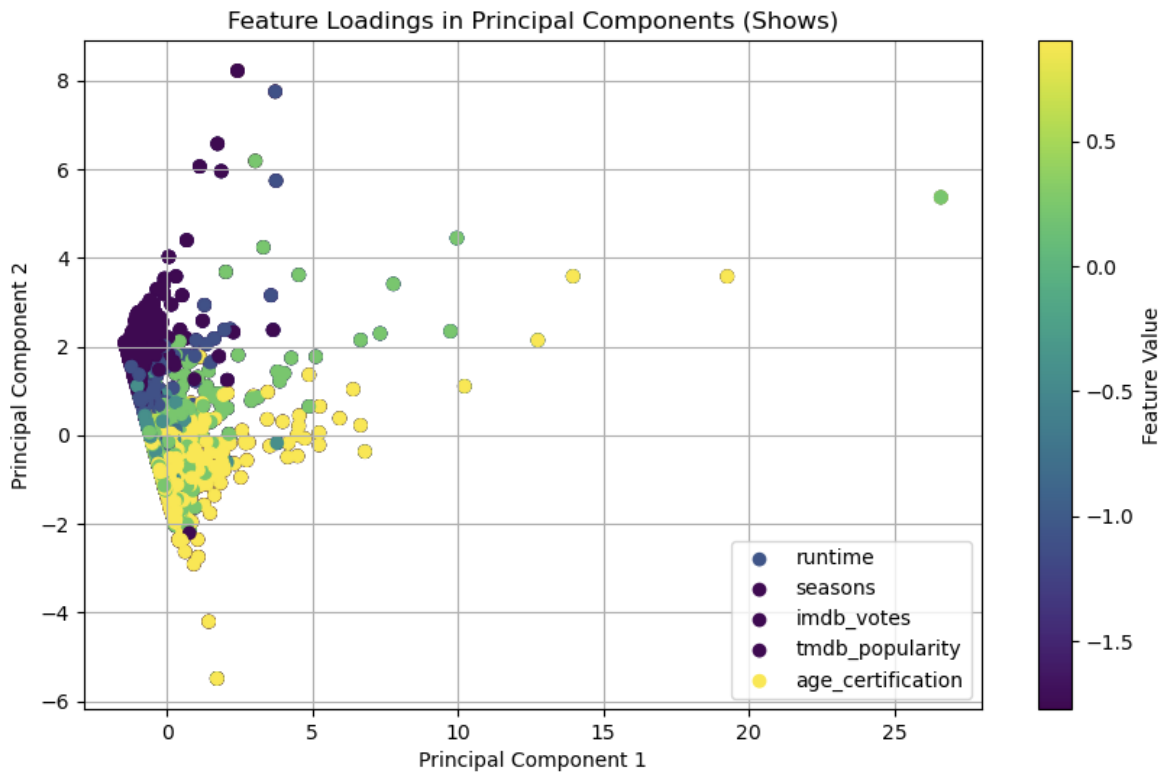
OUTPUT:

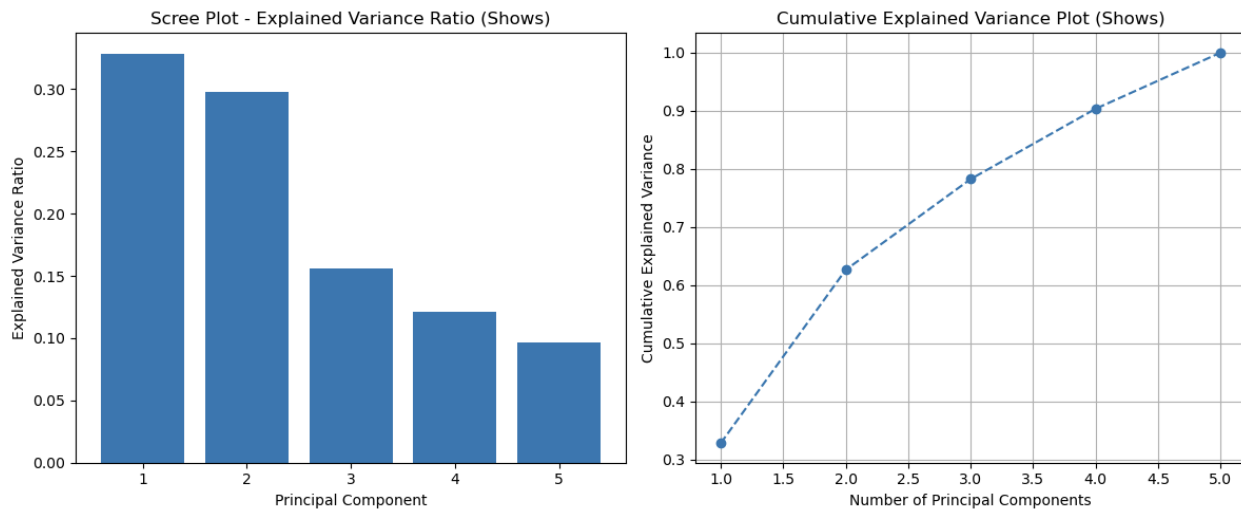
-----unsupervised movies model output----- Explained Variance Ratios for Principal Components (Movies): PC1: 0.3195 PC2: 0.2449 PC3: 0.2314 PC4: 0.2042 Cumulative Explained Variance: Components 1: 0.3195 Components 2: 0.5644 Components 3: 0.7958 Components 4: 1.0000	-----unsupervised shows model output----- Explained Variance Ratios for Principal Components: PC1: 0.3286 PC2: 0.2980 PC3: 0.1561 PC4: 0.1209 PC5: 0.0966 Cumulative Explained Variance: Components 1: 0.3286 Components 2: 0.6265 Components 3: 0.7826 Components 4: 0.9034 Components 5: 1.0000
Loadings of Original Features in First Two Principal Components:	Loadings of Original Features in First Two Principal Components:

runtime in PC1: 0.5504, PC2: -0.2185 imdb_votes in PC1: 0.6255, PC2: 0.0098 tmdb_popularity in PC1: 0.3265, PC2: 0.8916 age_certification in PC1: 0.4464, PC2: -0.3964	runtime in PC1: 0.2446, PC2: -0.5897 seasons in PC1: 0.2436, PC2: 0.4946 imdb_votes in PC1: 0.6472, PC2: 0.1341 tmdb_popularity in PC1: 0.6306, PC2: 0.1444 age_certification in PC1: 0.2536, PC2: -0.6073 imdb_votes in PC1: 0.6464, PC2: 0.1376 tmdb_popularity in PC1: 0.6298, PC2: 0.1477 age_certification in PC1: 0.2571, PC2: -0.6061
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Visualisations:







Findings and Analysis:

For movies, the principal component analysis (PCA) revealed that the first two components collectively explain approximately 56.44% of the variance. Notably, the primary determinant of popularity is IMDB votes, positively correlated with the first principal component (PC1), indicating that user engagement significantly influences a movie's success. Meanwhile, TMDB popularity plays a crucial role in the second principal component (PC2), highlighting the impact of online presence and marketing.

On the other hand, when analysing the shows, it shows that the first two components capture 62.5% variance approximately. PC1 is highly influenced by IMDB votes and TMDB popularity. This tells us that user feedback and online presence are the biggest factors for higher ratings. PC2, on the other hand, is characterised by factors such as runtime and age certification, emphasising the importance of content duration and suitability for specific audiences.

Model Limitations:

- PCA implies linear correlations between variables, which may not be true for all factors impacting popularity, thus leading to oversimplification.
- Loss of Interpretability: While PCA decreases dimensionality, it may make interpreting the meaning of the primary components in real-world terms more difficult.
- Feature Selection: Because the model is based on pre-selected features, it may ignore crucial elements that are not included in the study.

Possible Future Improvements:

Additional data, such as genre and cast information, could be added to this model in the future to capture more complex factors driving popularity. Furthermore, using data imputation techniques and investigating non-linear dimensionality reduction approaches other than PCA may improve the model's accuracy in forecasting movie and show success.

9. Conclusion

In this research report about factors influencing movie and show popularity, we have used multiple data pre-processing techniques, supervised regression, and unsupervised principal component analysis. There were some interesting findings in our pre-processed data analysis regarding clusters and placeholder values affecting them. Although the supervised regression models had limited predictive power, they still shed light on some critical relationships between features such as age_certification, runtime, imdb votes, and tmdb popularity. On the other hand, our unsupervised principal component analysis unravelled the importance of specific attributes that explain the variance of the dataset.

Going forward, researchers can reference this report to further investigate these relationships using more advanced statistical modelling techniques, most of which are very easily reproducible in Python or R. The use of a non-linear regression (e.g., multinomials, etc.) would serve as a better platform to analyse so many features.